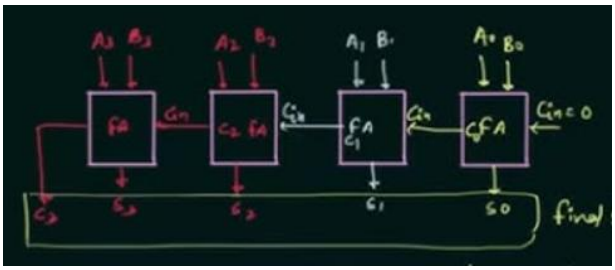
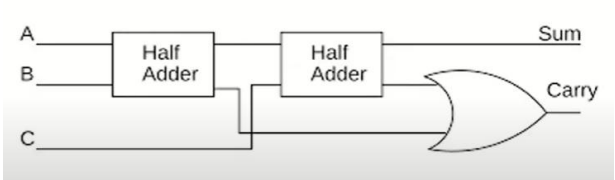
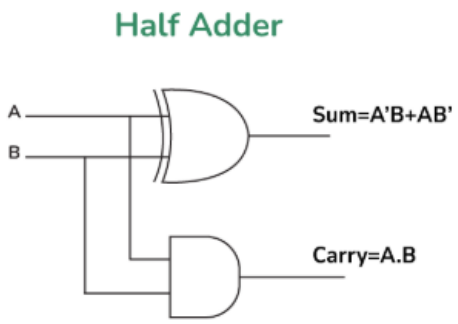
4bit Full Adder using halfAdder

Diagram:







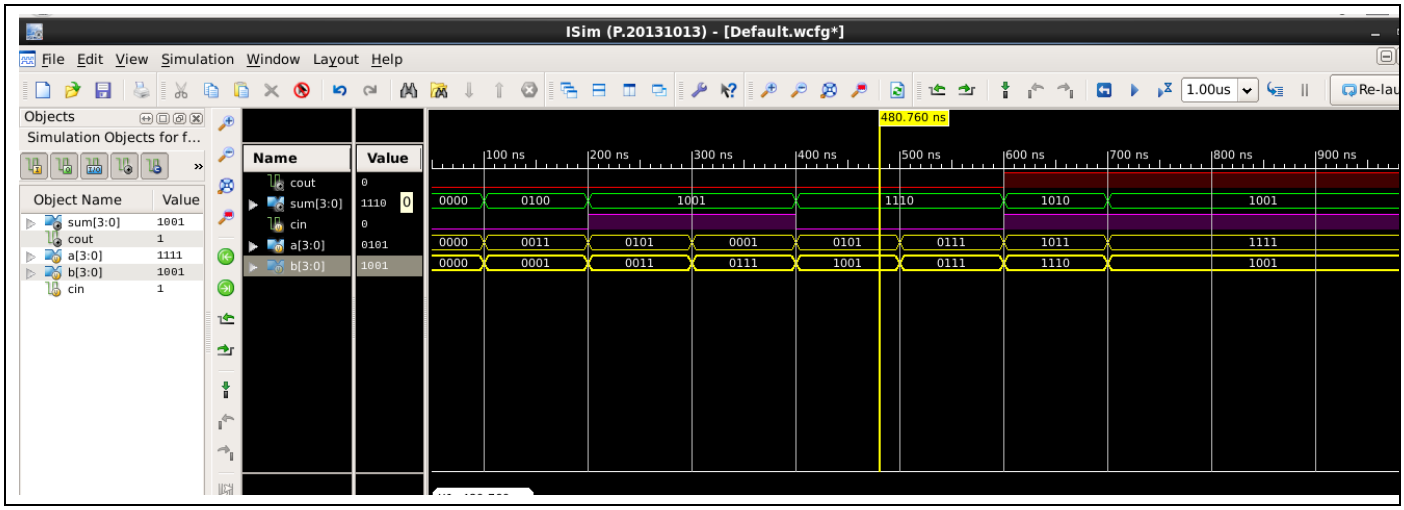
Verilog Code:

```
module half_adder (a,b,sum,carry);
input a,b;
output sum,carry;
assign sum = a^b;
assign carry = a&b;
endmodule

module fulladder (a, b, cin, sum, carry);
input a, b, cin;
output sum, carry;
wire w1, w2, w3;
half_adder h1(.a(a), .b(b), .sum(w1), .carry(w2));
half_adder h2(.a(w1), .b(cin), .sum(sum), .carry(w3));
or o1(carry, w3, w2);
endmodule

module 4bit_fulladder (input [3:0] a,b, input cin, output [3:0] sum, output cout);
wire c1, c2, c3;
fulladder fa0 (.a(a[0]), .b(b[0]), .cin(cin), .sum(sum[0]), .carry(c1));
fulladder fa1 (.a(a[1]), .b(b[1]), .cin(c1),   .sum(sum[1]), .carry(c2));
fulladder fa2 (.a(a[2]), .b(b[2]), .cin(c2),   .sum(sum[2]), .carry(c3));
fulladder fa3 (.a(a[3]), .b(b[3]), .cin(c3),   .sum(sum[3]), .carry(cout));
endmodule
```

Verilog Code tb:

```
module 4bit_fulladder_tb (input [3:0] a,b, input cin, output [3:0] sum, output cout);
reg [3:0] a,b;
reg cin;
wire [3:0] sum;
wire cout;
4bit_fulladder dut(.a(a), .b(b), .cin(cin), .sum(sum), .cout(cout));
initial begin
    a=4'b0000; b=4'b0000; cin=1'b0;
#100 a=4'b0011; b=4'b0001; cin=1'b0;
#100 a=4'b0101; b=4'b0011; cin=1'b1;
#100 a=4'b0001; b=4'b0111; cin=1'b1;
#100 a=4'b0101; b=4'b1001; cin=1'b0;
#100 a=4'b0111; b=4'b0111; cin=1'b0;
#100 a=4'b1011; b=4'b1110; cin=1'b1;
#100 a=4'b1111; b=4'b1001; cin=1'b1;
end
initial begin
$monitor($time, "a=%b, b=%b, cin=%b, sum=%b, cout=%b", a, b, cin, sum, cout);
end
endmodule
```

Simulation: